

List 6- Fin. Ele

► Consider the space $P_2(a, b) = \{v, v = \alpha_0 + \alpha_1 x + \alpha_2 x^2\}$. Compute the basis $\{\phi^1, \phi^2, \phi^3\}$ when we choose as degrees of freedom

$$\sigma_1(v) = v(a)$$

$$\sigma_2(v) = v\left(\frac{a+b}{2}\right)$$

$$\sigma_3(v) = v(b)$$

Considering that $\sigma_i(\phi^j) = \delta_{ij}$, we have
 $\sigma_1(\phi^1) = \phi^1(a) = 1$
 $\sigma_2(\phi^1) = \phi^1\left(\frac{a+b}{2}\right) = 0$
 $\sigma_3(\phi^1) = \phi^1(b) = 0$
 So $\frac{a+b}{2}$ and b are roots of ϕ^1 . So
 $\phi^1(x) = \theta_1\left(x - \frac{a+b}{2}\right)(x - b) =$

$$= \theta_1\left(x^2 - bx - \frac{a+b}{2}x + \frac{b(a+b)}{2}\right) = \theta_1\left(x^2 - \left(\frac{a+3b}{2}\right)x + \frac{ab+b^2}{2}\right)$$

As $\phi^1(a) = 1$, we have

$$\theta_1\left(a^2 - \left(\frac{a+3b}{2}\right)a + \frac{ab+b^2}{2}\right) = 1$$

$$a^2 - \frac{a^2}{2} - \frac{3ab}{2} + \frac{ab}{2} + \frac{b^2}{2} = \frac{a^2}{2} - ab + \frac{b^2}{2} = \frac{1}{2}(a^2 - 2ab + b^2) =$$

$$= \frac{(a-b)^2}{2} \quad \therefore \phi^1(x) = \frac{1}{(a-b)^2} \left(2x^2 - (a+3b)x + ab + b^2 \right)$$

$$\Rightarrow \theta_1 = \frac{2}{(a-b)^2}$$

Notice that ϕ^3 is symmetric to ϕ^1 . It's enough to replace a with b and b with a .

$$\phi^3(x) = \frac{1}{(a-b)^2} \left(2x^2 - (b+3a)x + ab + a^2 \right)$$

For ϕ^2 :

$$\left. \begin{aligned} \sigma_1(\phi^2) &= \phi^2(a) = 0 \\ \sigma_2(\phi^2) &= \phi^2\left(\frac{a+b}{2}\right) = 1 \\ \sigma_3(\phi^2) &= \phi^2(b) = 0 \end{aligned} \right\} \Rightarrow \phi^2(x) = \theta_2(x-a)(x-b) = \theta_2\left(x^2 - (a+b)x + ab\right)$$

$$\phi^2\left(\frac{a+b}{2}\right) = 1 \Rightarrow \theta_2\left(\left(\frac{a+b}{2}\right)^2 - (a+b)\left(\frac{a+b}{2}\right) + ab\right) = 1$$

$$\Rightarrow \theta_2\left((a+b)^2\left(\frac{1}{4} - \frac{1}{2}\right) + ab\right) = 1$$

$$\Rightarrow \theta_2\left(-\frac{1}{4}(a^2 + 2ab + b^2) + ab\right) = 1 \Rightarrow$$

$$\Rightarrow \theta_2\left(-\frac{a^2}{4} - \frac{1}{2}ab - \frac{b^2}{4} + \frac{2ab}{2}\right) = 1 \Rightarrow$$

$$\Rightarrow \frac{-\theta_2}{4}(a^2 - 2ab + b^2) = \frac{-\theta_2}{4}(a-b)^2 = 1 \Rightarrow$$

$$\Rightarrow \theta_2 = \frac{-4}{(a-b)^2}$$

$$\therefore \phi^2(x) = \frac{-4}{(a-b)^2} \left(x^2 - (a+b)x + ab \right)$$

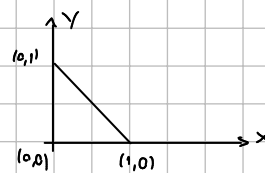
Therefore, the base functions are:

$$\phi^1(x) = \frac{1}{(a-b)^2} \left(2x^2 - (a+3b)x + ab + b^2 \right)$$

$$\phi^2(x) = \frac{-4}{(a-b)^2} \left(x^2 - (a+b)x + ab \right)$$

$$\phi^3(x) = \frac{1}{(a-b)^2} \left(2x^2 - (b+3a)x + ab + a^2 \right)$$

► Write the linear functions for the master triangle $\hat{K}$ having as vertices $((0, 0), (1, 0), (0, 1))$.



Letting $P_1 = (0, 0)$; $P_2 = (1, 0)$; $P_3 = (0, 1)$, we define the Dofs

$$\sigma_1(v) = v(P_1)$$

$$\sigma_2(v) = v(P_2)$$

$$\sigma_3(v) = v(P_3)$$

We want to find $\phi_1, \phi_2, \phi_3 \in P_1(\hat{K}) = \{v : v = \alpha + \beta x + \gamma y\}$ such that $\sigma_i(\phi_j) = \delta_{ij}$

$$\sigma_1(\phi_1) = \phi_1(P_1) = 1 \Rightarrow \alpha = 1$$

$$\sigma_2(\phi_1) = \phi_1(P_2) = 0 \Rightarrow \alpha + \beta = 0 \Rightarrow \beta = -1$$

$$\sigma_3(\phi_1) = \phi_1(P_3) = 0 \Rightarrow \alpha + \gamma = 0 \Rightarrow \gamma = -1$$

$$\therefore \phi_1(x, y) = 1 - (x + y)$$

$$\sigma_1(\phi_2) = \phi_2(P_1) = 0 \Rightarrow \alpha = 0$$

$$\sigma_2(\phi_2) = \phi_2(P_2) = 1 \Rightarrow \beta = 1$$

$$\sigma_3(\phi_2) = \phi_2(P_3) = 0 \Rightarrow \gamma = 0$$

$$\therefore \phi_2(x, y) = x$$

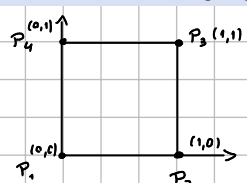
By symmetry, $\phi_3(x, y) = y$

$$\therefore \phi_1 = 1 - (x + y)$$

$$\phi_2 = x$$

$$\phi_3 = y$$

► Considering the bilinear functions $v(x, y) = \alpha_0 + \alpha_1 x + \alpha_2 y + \alpha_3 xy$ and the master quadrangle $\hat{K} = [-1, 1] \times [-1, 1]$. Write the basis functions considering as dofs the function values at the vertices (Lagrange).



Dofs:

$$\sigma_1(v) = v(P_1)$$

$$\sigma_2(v) = v(P_2)$$

$$\sigma_3(v) = v(P_3)$$

$$\sigma_4(v) = v(P_4)$$

We must have $\sigma_i(\psi_j) = \delta_{ij}$, with

$$\psi_j = \alpha_0^j + \alpha_1^j x + \alpha_2^j y + \alpha_3^j xy$$

ψ_1 :

$$\sigma_1(\psi_1) = \psi_1(P_1) = 1 \Rightarrow \alpha_0^1 = 1$$

$$\sigma_2(\psi_1) = \psi_1(P_2) = 0 \Rightarrow \alpha_0^1 + \alpha_1^1 = 0 \Rightarrow \alpha_1^1 = -1$$

$$\sigma_3(\psi_1) = \psi_1(P_3) = 0 \Rightarrow \alpha_0^1 + \alpha_1^1 + \alpha_2^1 + \alpha_3^1 = 0$$

$$\sigma_4(\psi_1) = \psi_1(P_4) = 0 \Rightarrow \alpha_2^1 = -1$$

$$\Rightarrow 1 - 2 + \alpha_3^1 = 0 \Rightarrow \alpha_3^1 = 1$$

$$\therefore \psi_1(x, y) = 1 - (x + y) + xy$$

ψ_2 :

$$\sigma_1(\psi_2) = \psi_2(P_1) = 0 \Rightarrow \alpha_0^2 = 0$$

$$\sigma_2(\psi_2) = \psi_2(P_2) = 1 \Rightarrow \alpha_0^2 + \alpha_1^2 = 1 \Rightarrow \alpha_1^2 = 1$$

$$\sigma_3(\psi_2) = \psi_2(P_3) = 0 \Rightarrow \alpha_0^2 + \alpha_1^2 + \alpha_2^2 = 0 \Rightarrow \alpha_2^2 = -1$$

$$\sigma_4(\psi_2) = \psi_2(P_4) = 0 \Rightarrow \alpha_0^2 + \alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 0 \Rightarrow \alpha_3^2 = -1$$

$$\psi_2(x, y) = x - xy$$

ψ_3 :

$$\psi_3(x, y) = y - xy, \text{ by symmetry}$$

ψ_4 :

$$\sigma_1(\psi_4) = \psi_4(P_1) = 0 \Rightarrow \alpha_0^4 = 0$$

$$\sigma_2(\psi_4) = \psi_4(P_2) = 0 \Rightarrow \alpha_0^4 + \alpha_1^4 = 0 \Rightarrow \alpha_1^4 = 0$$

$$\sigma_3(\psi_4) = \psi_4(P_3) = 0 \Rightarrow \alpha_0^4 = 0$$

$$\sigma_4(\psi_4) = \psi_4(P_4) = 1 \Rightarrow \alpha_3^4 = 1$$

$$\psi_4(x, y) = xy$$

► Noticing that the support of function ϕ^j in Figure 6.15 are all the elements sharing node j , what are the consequences for the matrix A with $A_{ij} = a(\phi^i, \phi^j)$, when choosing such space to compute an approximation to u ?

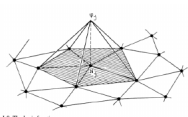


Figure 6.15: Hat function and an element-wise P_1 function on a triangular mesh.



The matrix A will have the maximum number of elements in a row equal to the max number of triangles shared by a node.

This makes the matrix A sparse.

► Show that Property (iii) in the definition of a finite element is equivalent to:

$$\sigma_i(p) = 0 \Leftrightarrow p = 0, \quad i = 1, \dots, n \quad (6.29)$$

Definition 6.5.1 (Ciarlet) A finite element in $\mathbb{R}^d$ (typically $d = 1, 2$ or 3) is a triplet (K, P_K, Σ_K) where

- (i) K is a closed bounded subset of $\mathbb{R}^d$ with a non-empty interior and Lipschitz boundary;
- (ii) P_K is a finite dimensional space of functions defined over K of dimension n ;
- (iii) Σ_K is a set of n linear functionals $\{\sigma_i\}_{i=1, \dots, n}$ such that for any real scalars $\alpha_i, i = 1, \dots, n$ there exist a unique function $p \in P_K$ that satisfies (unisolvence)

$$\sigma_i(p) = \alpha_i \quad (6.19)$$

$$\text{iii)} \Rightarrow (\sigma_i(p) = 0 \Leftrightarrow p = 0, \quad i = 1, \dots, n)$$

Notice that

$$\sigma_i(0) = \sigma_i(0+0) = 2\sigma_i(0) \Rightarrow \sigma_i(0) = 0, \quad \forall i$$

As the function is unique, $p=0 \in P_K$ is the only function that $\sigma_i(p)=0, \forall i$. So $\sigma_i(p)=0 \Rightarrow p=0$

If $p=0$, we already shown that $\sigma_i(p)=0, \forall i$

$$(\sigma_i(p)=0 \Leftrightarrow p=0, \forall i) \Rightarrow \text{iii)}$$

Take $\{\alpha_i\}_{i=1}^n$ scalars and $v, w \in P_K$ such that

$$\sigma_i(v) = \alpha_i, \quad \forall i$$

$$\sigma_i(w) = \alpha_i$$

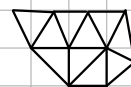
$$\text{So } \sigma_i(v) - \sigma_i(w) = \alpha_i - \alpha_i = 0 \Rightarrow \sigma_i(v-w) = 0, \quad \forall i$$

$$\Rightarrow v-w=0 \Rightarrow v=w$$

So iii) is valid.

► Which is the dimension of the space V_h given by Equation 6.18 associated to a mesh $\mathcal{T}_h$ made of triangles?

$$V_h = P_2(\mathcal{T}_h) = \{v \in C^0(\bar{\Omega}), v|_K \in P_2(K) \quad \forall K \in \mathcal{T}_h\} \quad (6.18)$$



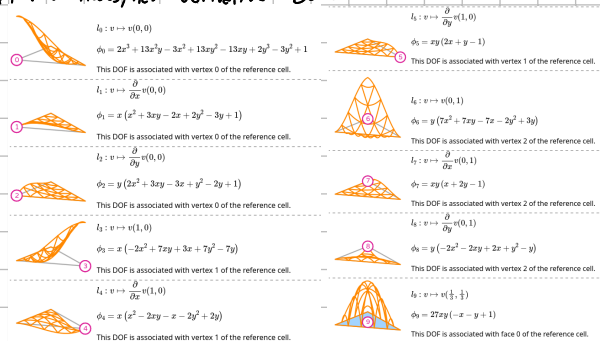
For each node we have a DoF. For each edge we have a DoF. This is because of the

continuity constraint

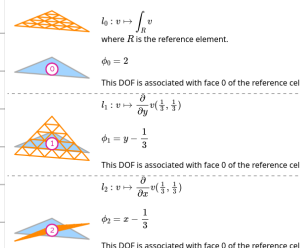
$$\therefore \dim(V_h) = N_v + N_{\text{edges}}$$

► Take a look at the finite element library that describes most of the known Finite Elements: [DeFElement](#). Use this library to plot the basis functions corresponding to 5 different finite elements of your choice in 2D, explaining the definitions of the chosen elements.

Hermite: Uses $P_3(K)$. Continuous and continuous derivative. 10 DoFs: 3 nodes and the barycenter. At the nodes, has derivative DoFs.



Degree 1 Taylor: Continuous. 3 DoFs at the center.



Vector Lagrange (Degree 1):

Continuous. 6 DoFs: 2 at each node. $\varphi_i: \hat{K} \rightarrow \mathbb{R}^2$

- $\mathcal{V}$ is spanned by: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} x \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix}, \begin{pmatrix} y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ y \end{pmatrix}$
- $\mathcal{L} = \{l_0, \dots, l_5\}$
- Functionals and basis functions:



$$l_0: v \mapsto v(0, 0) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_0 = \begin{pmatrix} -x - y + 1 \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 0 of the reference cell.



$$l_1: v \mapsto v(0, 0) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_1 = \begin{pmatrix} 0 \\ -x - y + 1 \end{pmatrix}$$

This DOF is associated with vertex 0 of the reference cell.



$$l_2: v \mapsto v(1, 0) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 1 of the reference cell.



$$l_3: v \mapsto v(1, 0) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_3 = \begin{pmatrix} 0 \\ x \end{pmatrix}$$

This DOF is associated with vertex 1 of the reference cell.



$$l_4: v \mapsto v(0, 1) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_4 = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 2 of the reference cell.



$$l_5: v \mapsto v(0, 1) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_5 = \begin{pmatrix} 0 \\ y \end{pmatrix}$$

This DOF is associated with vertex 2 of the reference cell.

- $\mathcal{V}$ is spanned by: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} x \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix}, \begin{pmatrix} y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ y \end{pmatrix}, \begin{pmatrix} xy(-x-y+1) \\ 0 \end{pmatrix}$
- $\mathcal{L} = \{l_0, \dots, l_5\}$
- Functionals and basis functions:



$$l_0: v \mapsto v(0, 0) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_0 = \begin{pmatrix} 9x^2y + 9xy^2 - 9xy - x - y + 1 \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 0 of the reference cell.



$$l_1: v \mapsto v(0, 0) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_1 = \begin{pmatrix} 0 \\ 9x^2y + 9xy^2 - 9xy - x - y + 1 \end{pmatrix}$$

This DOF is associated with vertex 0 of the reference cell.



$$l_2: v \mapsto v(1, 0) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} x(9xy + 9y^2 - 9y + 1) \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 1 of the reference cell.



$$l_3: v \mapsto v(1, 0) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_3 = \begin{pmatrix} 0 \\ x(9xy + 9y^2 - 9y + 1) \end{pmatrix}$$

This DOF is associated with vertex 1 of the reference cell.



$$l_4: v \mapsto v(0, 1) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_4 = \begin{pmatrix} y(9x^2 + 9xy - 9x + 1) \\ 0 \end{pmatrix}$$

This DOF is associated with vertex 2 of the reference cell.



$$l_5: v \mapsto v(0, 1) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\phi_5 = \begin{pmatrix} 0 \\ y(9x^2 + 9xy - 9x + 1) \end{pmatrix}$$

This DOF is associated with vertex 2 of the reference cell.



$$l_6: v \mapsto v(\frac{1}{2}, \frac{1}{2}) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\phi_6 = \begin{pmatrix} 27xy(-x-y+1) \\ 0 \end{pmatrix}$$

This DOF is associated with face 0 of the reference cell.



$$l_7: v \mapsto v(\frac{1}{2}, \frac{1}{2}) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

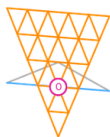
$$\phi_7 = \begin{pmatrix} 0 \\ 27xy(-x-y+1) \end{pmatrix}$$

This DOF is associated with face 0 of the reference cell.

Crouzeix-Raviart:

Discontinuous. 3 DoFs: 1 at the midpoint of each edge. Continuity only at those points

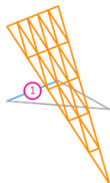
- $\mathcal{V}$ is spanned by: $1, x, y$
- $\mathcal{L} = \{l_0, \dots, l_2\}$
- Functionals and basis functions:



$$l_0: v \mapsto v(\frac{1}{2}, 0)$$

$$\phi_0 = 1 - 2y$$

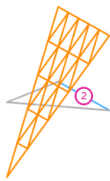
This DOF is associated with edge 0 of the reference cell.



$$l_1: v \mapsto v(0, \frac{1}{2})$$

$$\phi_1 = 1 - 2x$$

This DOF is associated with edge 1 of the reference cell.



$$l_2: v \mapsto v(\frac{1}{2}, \frac{1}{2})$$

$$\phi_2 = 2x + 2y - 1$$

This DOF is associated with edge 2 of the reference cell.

Vector bubble enriched Lagrange (Degree 1):

Continuous. 8 DoFs: 2 at each node (1 for each direction) and 2 at the center (1 for each direction)